

Supplementary Information for
*From signatures to populations: when does single-cell information
improve drug retrieval?*

This Supplementary Information provides additional methodological detail, robustness analyses and supporting experiments for the main text. It contains four Supplementary Notes and four Supplementary Tables. It carries no Extended Data figures: all 21 panels of the former Extended Data deck are now in the main figures. The supplementary analyses follow the same conceptual structure as the main text: they examine the information retained by population-level retrieval, the dependence of apparent retrieval gains on the evaluation criterion, and the conditions under which state-specific single-cell responses can become consequential for candidate ranking.

12 **Supplementary Note 1. Verification and transfer limits of HIR-Bench**

13 HIR-Bench was introduced as a controlled construction in which the conditions under which within-
14 population structure can alter candidate preference are known analytically. It is therefore used to verify
15 the logic of a preference-changing regime rather than to infer a biological phase boundary from data.

16 The generator satisfies its internal consistency checks on the full parameter grid. When the two sub-
17 populations have no conflict in candidate preference, the retrieval-flip rate is 0, as required by construc-
18 tion. Across the remaining parameter settings, preference changes occur near the analytical boundary

$$\alpha^* = \frac{B}{A + B},$$

19 and the median distance from the observed flip location to this boundary is 0.27. These checks con-
20 firm that the benchmark behaves according to its specified latent-welfare model.

21 We next asked whether the synthetic regime could be recognized from quantities observable at re-
22 trieval time. Observable features included the subpopulation-variance ratio, isotropy index and re-
23 sponse diversity of the query and candidate populations. Their performance is reported in the main
24 text. Features derived directly from the latent utility construction were analysed only as a controlled
25 comparison and are not available to a retrieval system.

26 The analytical boundary was also projected to the real-data retrieval tasks as a transfer test. This
27 projection did not generalize. The resulting real-data labels were strongly imbalanced (228 versus 11),
28 for which a constant majority-class predictor achieves 95.4% accuracy. The synthetic-regime projection
29 reached 61.1%, substantially below this baseline. The projection was necessarily fitted on the reduced
30 method-performance grid available for this analysis (144 instances, corresponding to 72 parameter cells),
31 rather than on the full HIR-Bench grid.

32 We therefore do not use the HIR-Bench boundary to classify biological datasets by whether pop-
33 ulation structure changes candidate preference. Its role is narrower: it establishes constructively that
34 within-population structure can become consequential for candidate choice, while the failed real-data
35 transfer shows that the corresponding regime cannot be inferred reliably from this synthetic boundary
36 alone.

37 **Supplementary Note 2. Additional diagnostics of when population information changes** 38 **ranking**

39 The main text separates the existence of state-specific perturbation responses from their recoverability
40 and from the downstream requirement that they alter candidate ordering. We performed several ad-
41 ditional analyses to ask whether simple observables could identify queries for which population-level
42 retrieval is more useful.

43 **Query identifiability.** In the SciPlex3 partial-observation tasks, query identifiability was quantified
44 from the two-state response population independently of the retrieval outcome. Across 543 evaluable
45 query–seed combinations, identifiability was positively associated with absolute mechanism-of-action
46 nDCG (Spearman $\rho = 0.17$, $p = 6.3 \times 10^{-5}$). The relationship was not monotonic, however: mean nDCG
47 across identifiability tertiles was 0.285, 0.240 and 0.444 from lowest to highest identifiability.

48 This comparison is additionally confounded by candidate-library composition. Queries in the
49 highest-identifiability tertile contained a median of six same-mechanism candidates, compared with
50 two in each of the lower tertiles, and a larger relevant candidate pool mechanically increases the
51 attainable nDCG. Even in the highest-identifiability tertile, the median silhouette score was only 0.038,
52 indicating weakly resolved two-state structure in these particular cell-line mixtures. These observations
53 therefore do not establish identifiability as a reliable predictor of population-level retrieval gain.

54 **Unsupervised recovery of constructed two-state structure.** The main text reports recoverability in
55 the constructed two-state mixture as a single supervised ceiling against the best unsupervised method
56 (Fig. 5e). Three further measurements characterise the same limit and are reported here rather than in
57 the figure.

58 First, across nine unsupervised clustering configurations, the median adjusted Rand index against
59 the known two-source labels ranged from 0.037 to 0.106. The configurations were k -means with $k = 2$
60 in the raw, 10-dimensional PCA, 50-dimensional PCA and response spaces, two-component Gaussian
61 mixtures in the raw and 50-dimensional PCA spaces, and k -means with k selected from 3, 4 and 5 by
62 silhouette, reported separately by which k was selected. The highest value, 0.106, was obtained when
63 silhouette selection chose $k = 4$. No configuration approached the value of 0.5 conventionally treated as
64 reliable recovery, so the low ceiling is not specific to the clustering method used in the main analysis.

65 Second, the weak separation is visible in the raw geometry rather than only in an accuracy score. Sil-
66houette scores computed independently for each perturbation-response population under a two-cluster
67 partition had a median of 0.034 in SciPlex3 ($n = 529$ populations) and 0.084 in Frangieh ($n = 231$);
68 in both datasets fewer than 0.5% of populations exceeded 0.1. Visible multimodality is therefore not a
69 prerequisite for, and should not be equated with, recoverable state-specific perturbation information.

70 Third, recovery is limited by state separation rather than by cell number. Averaging the k -means $k =$
71 2 adjusted Rand index over 20 seeds across a grid of source separation and cells per source, increasing the
72 cell budget from 25 to 400 cells per source changed recovery by at most 0.02 at fixed separation, whereas
73 increasing separation from 1.0 to 8.0 times the observed value raised it from 0.07 to 1.00. The separation
74 transform was defined from the full source populations before subsampling. Additional cells therefore
75 do not compensate for states that are close together, which is the same conclusion the supervised ceiling
76 reaches by a different route.

77 **Measured response divergence.** We next stratified the 600 mechanism-evaluable queries by the cosine
78 between their state-specific perturbation responses. Greater divergence was associated weakly with a
79 larger population-minus-mean mechanism-recovery difference (Spearman $\rho = 0.14$, permutation $p =$
80 2×10^{-4}), but the conditional advantage did not become convincingly positive. The most divergent
81 stratum reached a mean nDCG difference of approximately +0.02, with a bootstrap interval spanning
82 zero, and the direction reversed in one of the three SciPlex3 cell lines. Stratifying by query identifiability
83 instead yielded essentially no association with retrieval advantage.

84 These results distinguish differential response from decision relevance. Stronger state-specific re-
85 sponses may create the information required for population-aware retrieval, but they do not guarantee
86 that this information changes the ordering of candidate perturbations.

87 **Power of the two task-proximal comparisons in the highest-divergence stratum.** A null result invites
88 the objection that the comparison could not have detected an effect, so we computed achieved power
89 and the sample size required for 80% power directly, in the highest response-divergence quartile: the
90 stratum in which the pre-specified diagnostic itself predicts the largest population-level advantage, and
91 therefore the only stratum in which the objection has force. Two task-proximal criteria were compared
92 on the same queries. For minority-state coverage the observed mean gap was +0.0056 with a standard
93 deviation of 0.0135 across $n = 191$ queries, giving an achieved power of 1.00 and requiring only 45
94 queries for 80% power. For mechanism-of-action nDCG the observed mean gap was +0.0032 with a
95 standard deviation of 0.163 across $n = 143$ queries, giving an achieved power of 0.06 and requiring
96 20,844 queries for 80% power.

97 The two gaps differ by less than a factor of two; the two standard deviations differ by a factor of
98 twelve. The mechanism-recovery comparison is therefore uninformative at the sample size available,
99 and this is a property of that metric's variance rather than of the effect size. The correct reading is asym-
100 metric and we state it as such: the minority-state comparison is well powered and returns a small effect,
101 which is evidence of a small effect; the mechanism-recovery comparison is not adequately powered, so

its null is an absence of evidence rather than evidence of absence. The main text’s mechanism-recovery conclusion rests on the full 600-query set and on the shape of the whole distribution (Fig. 3a,b), not on this stratum.

These figures were reported in the main text until 2026-08-31, when the two panels carrying them were moved here to keep the main figure to the evidence that carries its argument.

Instability of thresholded dominance counts. We also examined whether the real-data tasks could be summarized by counting queries whose population-level gain exceeded a fixed threshold. Such counts were unstable under resampling. Using a gain threshold of $+0.01$, only 2 of the 10 initially threshold-crossing cross-line tasks remained above threshold in at least 8 of 10 seed resamples. Conversely, control drugs that did not cross the threshold in the recorded run crossed it in between 1 and 4 of 10 resamples.

The two most stable examples were A549→MCF7 Abexinostat ($+0.090 \pm 0.013$ across resamples) and Belinostat ($+0.081 \pm 0.027$), both HDAC inhibitors and both above threshold in all 10 resamples. With only two perturbations from a single drug class and one cell-line pair, these examples are treated as hypotheses rather than evidence for a general regime in which population structure changes candidate preference.

For this reason, the main text reports continuous retrieval differences and stratified analyses rather than thresholded winner counts.

Two checks on the diagnostic’s decline verdicts. The pre-specified diagnostic declines 133 queries as “mean or no call” and a further 11 as “mean sufficient”. The 11 mean-sufficient queries are excluded from the main comparison because their regret reduction is exactly 0.000 by construction; pooling them into the non-recommended group changes nothing, giving a difference of $+0.005$ with a Mann–Whitney $p = 0.81$ and an interval covering zero.

Pooling the two decline verdicts into one group returns $p = 0.09$, which reads as an absence of effect but is two opposite effects cancelling: the 133 “mean or no call” queries are more divergent than the queries the diagnostic recommends, while the 11 “mean sufficient” queries lie below the fifth percentile of both groups, where the diagnostic is correct. A distributional test on the same pooling detects the difference the location test misses (Kolmogorov–Smirnov $D = 0.154$, $p = 0.007$).

Supplementary Note 3. Perturbation-predictor diagnostics

The predictor analyses in this study address a specific question rather than comparing perturbation-prediction models generally: whether a predicted response population contains state-specific perturbation effects that could be exploited by a downstream population-aware retriever.

Additive response models. For a predictor that applies the same perturbation effect δ_d to every cell in a context,

$$\hat{x}_{i,d} = x_i + \delta_d,$$

the response of any two cellular states is identical after subtraction of their corresponding controls,

$$r_{d,0} = r_{d,1} = \delta_d.$$

Consequently,

$$\cos(r_{d,0}, r_{d,1}) = 1.$$

137 This result is independent of the heterogeneity of the starting control population. An additive pre-
138 dictor can therefore generate a visually heterogeneous predicted population while containing no state-
139 specific variation in the perturbation effect itself.

140 The average-effect and linear-latent predictors used in the main analysis satisfy this additive limit by
141 construction when the perturbation response is referenced to the model-consistent baseline. Their role
142 is to establish this limiting case, not to represent the performance of modern perturbation-prediction
143 models as a class.

144 **Two reference baselines, and why both are reported.** A predicted population’s induced response can
145 be referred to two different controls, and the two answer different questions. Referring it to the model’s
146 own zero-effect baseline, which is its reconstruction of the control population whenever the model de-
147 codes and the control cells themselves when it does not, cancels the reconstruction bias and leaves the
148 divergence the model itself generates. The additivity result above pins this quantity to exactly 1 for any
149 predictor of the form $\hat{x}_{i,d} = g(x_i) + \delta_d$ with a fixed g , covering both the gene-space case $g = \text{id}$ and the la-
150 tent case in which g is the model’s own reconstruction, and it is asserted rather than fitted. Referring it to
151 the real control cells is instead what a retrieval scorer does, and the resulting cosine contains the model’s
152 reconstruction bias, which differs between cellular states and is not a perturbation response. Across the
153 same 30 cell-line and drug combinations, the average-effect, linear-latent, scGen and optimal-transport
154 predictors give 1.000, 1.000, 0.972 and 0.900 against the model baseline, and 1.000, 0.190, 0.404 and 0.380
155 against the real control; real treated cells give 0.205, and have no model baseline.

156 The two columns are informative together. Lower cosine means more state-specific divergence. The
157 average-effect predictor adds its effect directly in gene space and has no reconstruction step, so its two
158 references coincide at 1.000, which is the correct answer for such a model rather than a defect of the
159 measurement. The other three pass their output through a decoder whose reconstruction error differs
160 between cellular states, and that error enters the real-control cosine as if it were a perturbation effect. The
161 linear-latent model is the clearest case: it generates exactly zero state-specific divergence, yet measured
162 the way a scorer measures it it reads 0.190, marginally more divergent than the 0.205 of the real cells
163 it replaces. A distributional score run on such a predicted population is therefore partly scoring the
164 decoder. This is a second reason, independent of transfer accuracy, why predicted candidates do not
165 provide a clean test of whether population structure improves retrieval.

166 **scGen.** A deployed scGen model was evaluated using the same within-context cellular-state partition
167 and state-specific induced-response definition as the other predictors. Across the same 30 cell-line and
168 drug combinations, its mean induced-response cosine against its own reconstruction baseline was 0.972,
169 close to the additive limit of 1.000, and 0.404 against the real control. Thus, in this experiment scGen
170 preserved substantially less state dependence in the perturbation effect than the 0.205 observed in real
171 treated cells, under either reference.

172 The model is the published `scgen.SCGEN` implementation on `scvi-tools`, trained on the SciPlex3
173 tensor with the perturbation label as batch key and the cell line as label key, 30 latent dimensions, batch
174 size 512, at most 40 epochs with early stopping after 15 epochs without improvement, and a fixed seed.
175 Predictions use scGen’s own latent arithmetic: one latent delta per perturbation, taken as the mean latent
176 of treated cells minus the mean latent of control cells, added to the encoded control mean and decoded
177 (Methods).

178 **Non-parametric optimal-transport mapping.** We additionally evaluated a non-parametric optimal-
179 transport map as a predictor that is not constrained to apply a shared additive shift to all cells. Under
180 the within-context differential-response analysis, the optimal-transport predictions reached an induced-
181 response cosine of approximately 0.900 ± 0.107 against their own reconstruction baseline, moving away
182 from the additive limit but remaining much more aligned than the observed responses.

183 A latent-space implementation passed the corresponding perturbation-learning diagnostic (mean-
184 delta agreement 0.352), whereas the same construction applied directly in gene space showed substan-
185 tially weaker perturbation learning (0.057). The latent-space implementation was therefore used for the
186 differential-response comparison.

187 The map is an entropic (Sinkhorn) coupling between control and treated cells with uniform
188 marginals, regularization 0.05, at most 500 iterations and a cost matrix rescaled by its maximum, solved
189 in a 30-component PCA latent fitted unsupervised on at most 20,000 cells. Each fitting context is centred
190 on its own control mean, and the query context is excluded from the fit. A barycentric projection gives
191 a per-source-cell displacement, extended to the query's real control cells by nearest-source matching in
192 the same centred space and decoded through the PCA inverse transform (Methods).

193 Opening this state-specific response channel was not sufficient to provide a clean retrieval test. When
194 the same optimal-transport map was used to generate candidates in the cross-context predict-then-rank
195 setting, Hit@1 was approximately 0.05 under each retrieval rule, compared with 0.88 for the additive
196 average-effect candidate source. The optimal-transport predictions were therefore too inaccurate in that
197 transfer setting to test whether greater state dependence itself improves retrieval.

198 Taken together, these analyses leave a specific unresolved experiment. A decisive test requires a
199 predictor that both generalizes accurately to the target cellular context and reproduces the state-specific
200 perturbation responses present in real cells. No predictor evaluated here satisfied both requirements
201 simultaneously.

202 **Supplementary Note 4. Robustness of the natural-heterogeneity and large-scale analyses**

203 The main text uses patient-derived glioblastoma and Tahoe-100M to ask whether the information re-
204 quirements identified in the controlled experiments are also present when cellular heterogeneity is not
205 created by combining experimentally distinct source populations. These datasets are not additional
206 retrieval benchmarks. They measure differential response, recoverability and the extent to which state-
207 specific response ordering leaves headroom for population structure to alter candidate preference.

208 **Robustness of glioblastoma compartment assignment.** The ZhaoSims2021 dataset does not provide
209 resolved cell-type labels for the analysed cells, so malignant, tumour-associated myeloid and oligoden-
210 drocyte compartments were assigned from canonical marker scores. The primary analysis required both
211 an absolute marker-expression threshold of 0.25 and a z-score margin of 0.25 over the second-highest
212 compartment score. Cells failing these criteria were excluded rather than forced into a compartment,
213 resulting in 39.9% of cells being labelled unassigned.

214 The results were stable when both thresholds were varied jointly over

$$\{0.10, 0.25, 0.40, 0.50\}.$$

215 Across this range, the excluded fraction varied from 29% to 58%, while the supervised drug-response
216 ceiling remained between 0.915 and 0.923, the best unsupervised recovery between 0.77 and 0.80, and
217 the paired recoverability gap between approximately 0.11 and 0.13. The median cosine between
218 malignant- and myeloid-specific perturbation-response directions remained between 0.54 and 0.59.
219 Thus, the natural-tissue conclusions are not specific to the primary 0.25 assignment threshold.

220 **Patient-level uncertainty.** The natural-tissue recoverability analysis contains 36 within-patient drug-
221 pair splits across four patients and two compartments, with 30 of the 36 splits contributed by a single
222 patient. Drug-pair splits were therefore not treated as independent replicates. Patient-level cluster boot-
223 strap resampling was used to quantify uncertainty, and the analysis was repeated after removing the
224 dominant patient.

225 The reported recoverability gap is computed within each split before aggregation,

$$G_j = A_{\text{sup}}^{(j)} - A_{\text{unsup}}^{(j)},$$

rather than as the difference between the two marginal medians. The primary paired median is +0.117; subtracting the marginal medians would instead give +0.147. Across 5,000 patient-cluster bootstrap resamples, the natural-minus-constructed difference in recoverability gap was +0.109 with an interval of [+0.052, +0.129], and no bootstrap draw was at or below zero. Because only four patient clusters contribute to this calculation, the interval is interpreted as supportive rather than as a precise population-level uncertainty estimate.

The gap also survives two checks that remove the two obvious ways it could be an artefact of aggregation. Fixing every probe-cluster pair, so that the same pair is scored in both arms, gives +0.122 in tissue against -0.002 in the constructed arm. Removing the patient that contributes 30 of the 36 splits gives +0.123, with a [+0.098, +0.138] interval.

Large-scale measurement in Tahoe-100M. Tahoe-100M plate 3 provides a substantially larger setting in which heterogeneity occurs within cell lines rather than through constructed mixtures. Across 3,630 evaluable drug-cell-line conditions, the median cosine between the two state-specific response directions was 0.739 (interquartile range 0.600–0.836). Fewer than 0.11% of conditions were as divergent as the constructed mixtures used in the controlled experiments.

This estimate is not driven materially by the minimum-cell filter. Conditions were excluded when one of the required state-specific treated or control arms contained fewer than 50 cells, which can preferentially remove strong cell-cycle-arresting perturbations. Across drugs, coverage was only weakly associated with induced-response cosine (Spearman $\rho = 0.077$), and restricting the analysis to well-covered compounds changed the median from 0.739 to 0.740.

Recoverability remained measurable at scale. Across 960 sampled drug pairs and 48 cell lines, the supervised ceiling was 0.853 and the best unsupervised procedure reached 0.611, giving a median paired gap of 0.157. The gap was positive in all 48 evaluable cell lines. Individual unsupervised methods gave similar results: *k*-means 0.592, Gaussian mixtures 0.582 and Leiden 0.563.

State-specific ordering and decision-relevance headroom. We finally compared relative drug-response ordering between non-overlapping cellular states. Under the cell-cycle partition, the median disjoint Spearman correlation between the two drug-drug similarity structures was 0.841 across 44 evaluable contexts. Under the independently control-derived two-state partition, the corresponding median was 0.778 across 45 contexts.

These high correlations indicate that relative drug ordering is often substantially preserved between cellular states even when the states respond differently. The pattern is not universal: under the control-derived partition, 2 of 45 contexts had a correlation below 0.5, with a minimum of 0.236. Such contexts leave greater headroom for state-specific response information to alter candidate ordering, but this statistic alone does not establish that a population-level retrieval method would improve the selected candidate.

The two state definitions agree only moderately in their ranking of contexts (Spearman $\rho = 0.466$), reinforcing that apparent decision-relevance headroom depends on how cellular states are defined. Both partitions were specified independently of downstream retrieval performance, and neither was selected after observing which produced the larger effect.

Tahoe-100M therefore supports the same separation of concepts observed elsewhere in the study: state-specific responses can exist, and response identity can remain recoverable, while much of the relative candidate ordering is still preserved across cellular states. No retrieval algorithm was run on Tahoe-100M, so these analyses constrain the conditions under which population-level retrieval could add value rather than measuring such an advantage directly.

Supplementary Table 1. Data resources, preprocessing and analytical roles.

Resource	Scale	Perturbation/context structure	Representation and controls	Role in this study
SciPlex3	276,325 cells; 2,000 genes	188 compounds shared across A549, K562 and MCF7	Log-transformed 2,000-gene representation; perturbation responses referenced to matched controls	Primary observed-population drug-retrieval setting; controlled within- and cross-context constructions; source of the GDSC2-linked transcriptional rankings
Frangieh Perturb-CITE-seq	218,023 RNA-profiled cells; 2,000 genes; 20 retained surface proteins	239 CRISPR knock-outs across Control, IFN γ and Co-culture conditions	RNA and protein modalities joined by cell barcode; responses referenced to condition-matched controls	Independent genetic-perturbation retrieval setting and matched RNA-protein evaluator-shape experiment; not treated as a drug-retrieval benchmark
CD34+ HSPC	33,984 cells; 2,000 genes	36 compounds	Log-transformed expression; DMSO-referenced perturbation responses	Negative-control retrieval setting for testing whether population-level information consistently improves ranking
ZhaoSims2021 glioblastoma	10 patients; 6 compounds	Acute-slice culture and surgical biopsy with co-existing malignant, myeloid and oligodendrocyte compartments	Counts normalized to 10 ⁴ per cell, log1p-transformed and reduced to 2,000 highly variable genes; state-specific responses referenced within patient and compartment	Natural-tissue analysis of differential response, recoverability and state-specific response ordering; no retrieval benchmark is constructed
Tahoe-100M plate 3	4,158,278 cells; 2,304 genes	50 cell lines $\times$ 93 compounds at one concentration, with DMSO_TF controls	Released 2,304-gene log1p representation used without refitting; state-specific responses referenced to matched state controls	Large-scale measurement of differential response, recoverability and state-ordering agreement under intrinsic within-cell-line heterogeneity; no retrieval is run
GDSC2	286 compounds across 969 cell lines in the retained release	35 compounds matched to the SciPlex3 panel; A549, K562 and MCF7 excluded from external profiles	Fitted dose-response AUC; primary functional similarity uses cell-line-centred cross-line response profiles	External functional evaluation of fixed SciPlex3 RNA-derived rankings
HIR-Bench	13,440 synthetic instances	672 parameter configurations $\times$ 20 seeds in a two-subpopulation latent-welfare model	Synthetic observed, predicted-mean and predicted-structure information conditions	Controlled demonstration that population structure can alter candidate preference; not interpreted as a biological generative model

The resources above answer distinct questions and are not pooled as independent replicates of a single benchmark. Dataset-specific cell-number thresholds, subsampling rules and statistical units are given in the corresponding Methods sections.

Supplementary Table 2. Retrieval scores, task constructions and evaluation criteria.

(a) Retrieval scores.

Score	Input representation	Ranking rule	Role
Mean cosine	Control-referenced mean perturbation signatures	Descending cosine similarity between query and candidate mean-response vectors	Primary mean-signature baseline
CMap-style cosine	Same control-referenced mean differential-expression vectors as mean cosine	Same cosine operation as the implemented mean-cosine baseline	Signature-retrieval reference; numerically identical to mean cosine in this implementation
WTCS	Mean-collapsed differential-expression response	Rank-based weighted connectivity / enrichment score	Canonical rank-based signature baseline; related to but not algebraically identical to mean cosine
Energy distance	Empirical control-referenced single-cell response populations	Candidates ranked by increasing energy distance to the query population	Primary population-level retrieval score
RBF-MMD	Empirical response populations	Candidates ranked by increasing maximum mean discrepancy under an RBF kernel	Alternative distribution-sensitive score
Sliced Wasserstein	Empirical response populations	Candidates ranked by increasing sliced Wasserstein distance using 128 random projections	Alternative distribution-sensitive score
Subpopulation coverage	Response populations decomposed into response components	Component-level distances aggregated with temperature β ; increasing β gives progressively greater weight to the least well-matched response component	Population-sensitive score used for the partial-observation response-matching analyses

The five configured population-level scoring variants are, verbatim from the experiment registry, `energy`, `mmd`, `sliced_wasserstein`, `coverage_mean` and `coverage_worst`. The four score families become five configurations because subpopulation-coverage contributes two: coverage-mean is the $\beta \rightarrow 0$ limit of the aggregate and coverage-worst the $\beta \rightarrow \infty$ limit, both with energy distance as the per-component base metric. They are compared against `mean_cosine`, `mean_12`, `cmap_match` and `cmap_reverse`; `cmap_match` is numerically identical to `mean_cosine` on control-referenced mean vectors, and `cmap_reverse` is its negation.

Supplementary Table 2 continued. (b) Retrieval task constructions.

Task	Query and candidate construction	Candidate evaluation	Statistical unit
Observed-population retrieval over heterogeneous queries	A heterogeneous query population combines two response modes at weight α ; the library holds a held-out re-mixture at the same weight, the two single-mode populations, and 40 distractors, giving a 43-candidate library	Hit@1; only the both-mode candidate is relevant, so a single-mode population counts as a miss even when it carries one of the query's two response modes	Retrieval query; headline values macro-averaged over seven task-setting combinations
Controlled heterogeneous queries	Two response modes are combined at mixing weight α ; controlled SciPlex3 analyses use HDAC- and JAK-associated response sources	Source identification and response-matching criteria	Query / task setting
Partial-observation retrieval	Only the response information available under the controlled partial-observation protocol is exposed to the retriever; population and mean methods rank the identical candidate library once	The resulting fixed ranking is evaluated under response matching, MoA recovery and minority-state coverage	765 total queries
Mechanism-evaluable subset	Same partial-observation rankings; restricted to queries for which the query mechanism remains represented among candidates	MoA-nDCG	600 queries; 165 excluded because MoA recovery is undefined
Real-data minority-state projection	Top-ranked population and mean candidates evaluated on 239 task instances spanning constructed cross-line, within-line, Frangieh, CD34+ and predicted-population settings	Difference in minority-state coverage	Task-level difference; not interpreted as 239 independent biological replicates
GDSC2-linked retrieval	Held-out SciPlex3 compound at 10 μ M used as query; remaining matched compounds form a 34–35 candidate library	Spearman agreement between RNA-derived ranking and external GDSC2 functional-similarity ordering	Per-query correlation summarized within SciPlex3 cell line

Supplementary Table 2 continued. (c) Evaluation criteria applied to fixed rankings.

Criterion	Definition	Interpretation
Hit@1	Indicator that the both-mode candidate, the held-out re-mixture at the query’s own weight, is ranked first	Recovery of the query’s response composition, not of its perturbation identity alone
Response-matching regret	Energy-based welfare regret against the latent per-cell objective used to construct the controlled partial-observation task; reported as regret reduction relative to mean retrieval	Response fidelity; intentionally close to the population information used by the retriever
MoA-nDCG	Normalized discounted cumulative gain rewarding candidates that share the query perturbation’s annotated mechanism	Task-proximal biological criterion; undefined when the relevant mechanism is absent from the candidate library
Minority-state coverage	Measures whether the selected candidate better preserves the response of the less-represented cellular state	Task-proximal measure of coverage of heterogeneous response structure
GDSC2 functional similarity	Spearman correlation between cell-line-centred cross-cell-line AUC profiles of the query and candidate drugs	Externally measured functional similarity; not absolute potency
Protein mean evaluator	Cosine similarity between control-referenced mean protein-response vectors	External evaluator preserving only mean protein response
Protein population evaluator	Negative energy distance between control-referenced per-cell protein-response populations	External evaluator retaining cell-to-cell protein-response structure
Response-magnitude matching	$-\ \bar{d}_q\ - \ \bar{d}_c\ $	Query-dependent scalar control that uses no within-population information

Query states s are obtained by k -means with $k = 2$ on the query cells, the minority state being the smaller cluster. A candidate d with response population P_d receives the welfare $U(d) = \text{agg}_s(-\mathcal{E}(P_d, Q_s))$, where $\mathcal{E}$ is the energy distance and the aggregator is the minimum over states in the worst-case form and the mean in the mean form. The response-matching regret of a selected candidate $\hat{d}$ is $\text{Regret}(\hat{d}) = \max_{d \in \mathcal{L}} U(d) - U(\hat{d}) \geq 0$, taken over the observed library $\mathcal{L}$, so the reference is the best available candidate rather than an unobservable optimum. Minority-state coverage is $\frac{1}{2}(1 + \cos(\bar{P}_d, \bar{Q}_{\min}))$, the cosine between the candidate’s mean response and the mean response of the query’s minority state rescaled to $[0, 1]$. The same definitions are given in the main Methods.

Supplementary Table 3. Perturbation-predictor implementations and their role in the differential-response analysis.

Predictor	Model form	Predicted-population construction	Use in this study	Interpretation / limitation
Average-effect	Per-drug mean perturbation delta	Predicted delta applied to the real control cells of the query context	Predict-then-rank candidate source and within-context differential-response reference	Additive by construction; state-specific induced-response cosine equals 1
Nearest-neighbour	Transfers the perturbation effect from the training context with the most similar untreated control state	Transferred effect applied to real query-context control cells	Predict-then-rank candidate source	Tests simple context transfer rather than a general virtual-cell architecture
Linear-latent	Additive latent arithmetic in an encoded representation followed by linear decoding	Observed control cells encoded, shifted in latent space and decoded individually	Predict-then-rank candidate source and additive latent reference	Can preserve baseline cellular heterogeneity while remaining additive in perturbation effect
scGen	Variational-autoencoder perturbation model with latent arithmetic	Evaluated on the same within-context state partition and state-specific response definition used for other predictors	Additional differential-response diagnostic reported in the main text	Mean induced-response cosine 0.972 over 30 cell-line and drug combinations
Optimal-transport map	Non-parametric mapping that permits cell-dependent displacements	Control cells mapped individually to predicted treated states	Additional non-additive differential-response diagnostic; also tested as a candidate-response source	Within-context induced-response cosine approximately 0.900 ± 0.107 against the map's own reconstruction baseline and 0.380 against the real control; cross-context candidate predictions were too inaccurate for a clean retrieval test

Supplementary Table 3 continued. Implementation details required for reproducibility.

Predictor	Required final specification
Average-effect	Drug-effect estimation set, context handling, cell cap, and exact construction used in the reported predict-then-rank run
Nearest-neighbour	Control-state similarity metric, training contexts available to each query, tie handling and transferred-response construction
Linear-latent	Encoder dimensionality, fitting set, perturbation-delta estimation, decoder definition and population-synthesis procedure
scGen	Published <code>scgen.SCGEN</code> on <code>scvi-tools</code> ; trained on the SciPlex3 tensor, batch key = perturbation, label key = cell line; 30 latent dimensions; batch size 512; at most 40 epochs with early stopping after 15 epochs without improvement; fixed seed; per-perturbation latent delta added to the encoded control mean and decoded
Optimal transport	Entropic (Sinkhorn) coupling in a 30-component PCA latent; cost rescaled by its maximum; uniform marginals; regularization 0.05; at most 500 iterations; fitted per non-query context centred on that context’s own control mean, with the query context excluded; barycentric projection extended to the query’s control cells by nearest-source matching and decoded by the PCA inverse transform

Predictors are used as candidate-response sources or information diagnostics rather than as a benchmark of perturbation-prediction architectures. The reported predict-then-rank experiment uses predictors fitted on the full SciPlex3 matrix and is therefore interpreted as an in-sample ceiling, not as a cross-context generalization benchmark.

Supplementary Table 4. Sensitivity of the GDSC2 functional-similarity evaluation to response-profile construction.

The primary external functional criterion compares drug–drug response profiles after removing the shared cell-line sensitivity effect. For cell line l and drug d ,

$$\widetilde{\text{AUC}}_{d,l} = \text{AUC}_{d,l} - \frac{1}{|\mathcal{D}_l|} \sum_{d' \in \mathcal{D}_l} \text{AUC}_{d',l},$$

where $\mathcal{D}_l$ contains all GDSC2 compounds measured in that cell line. Drug–drug functional similarity is then the Spearman correlation between centred cross-cell-line response profiles. The table compares this primary construction with the corresponding uncentred AUC-profile similarity.

RNA-derived ranking rule	Cell-line-centred functional similarity	Uncentred functional similarity
Energy distance	+0.276	+0.0967
Mean cosine, control referenced	+0.083	+0.138
Mean cosine, treated mean only	+0.241	+0.092
Response-magnitude matching	+0.232	+0.217
Candidate magnitude only	−0.330	+0.135
Potency matching diagnostic	+0.399	+0.290
Energy, magnitude-adjusted partial rank correlation	+0.097	n/a

Values are median per-query Spearman correlations over 103 leave-one-drug-out queries across the three SciPlex3 cell lines. The primary centred functional criterion is constructed from GDSC2 AUC profiles after subtracting each cell line’s mean response across all 286 compounds in the retained release. Drug pairs sharing fewer than 100 cell lines are excluded; 591 usable drug pairs remain, with a median of 909 shared cell lines per pair.

The change between centred and uncentred constructions shows that external evaluation is itself sensitive to how the functional measurement is summarized. Under the centred criterion, energy retrieval exceeds the control-referenced mean-signature baseline, but much of this association is also reproduced by response-magnitude matching. Under the uncentred criterion, the ordering between energy and the mean-signature baseline reverses. Neither construction is interpreted as a direct measure of therapeutic utility.